Iter1.1 Problem Definition

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Problem Statement

Core Problem

University students in Singapore have access to countless productivity apps, but they're still struggling with procrastination and cannot be consistent with their habits. Basically there is a gap between what they plan to do and what they actually get done. This isn't just involving grades, it's causing real stress, wasting time they could be using to earn money or actually relax, and leaving them feeling behind.

Problem Details

1. **The Willpower Deficit**

Most productivity apps assume you already have self-discipline, which is exactly what procrastinating students don't have. When you're tracking goals in a private app and you skip a day, nothing happens or the consequences aren't something that truly affects the user (e.g. the tree is not growing). No one else also knows, no one cares. There are literally zero stakes, which makes it way too easy to hit snooze on your responsibilities when you're tired or distracted.

Our survey backs this up (from the project proposal): 85% of respondents struggle to stay consistent with their habits, and they're mostly motivated by external pressure like deadlines and grades rather than their own internal systems.

2. **The Temporal Disconnect**

Students have a hard time connecting what they do today with rewards that are months away. Studying for an exam that's three months out feels abstract and far away, but scrolling TikTok gives you instant gratification and dopamine. This is what behavioral economists call "temporal discounting" which basically means your brain is terrible at valuing future rewards over immediate pleasure.

The real damage here is the last-minute cramming that follows. Students pull all-nighters, barely retain what they learned, and mess up their sleep schedules. It's a vicious cycle that hurts both their performance and their health.

3. **The Pressure Paradox**

Singapore's academic environment is intense. Students face pressure from tough curriculums, competitive grading, and high living costs. But the weird thing is that sometimes all that pressure makes them procrastinate more, not less. When you're stressed, it's easier to avoid your work by scrolling social media or gaming than to actually face the mountain of tasks ahead.

There's also an opportunity cost here. Every hour spent procrastinating is an hour that could've been spent on part-time work (making money), building career skills, or getting proper rest to avoid burnout.

4. **The Social Isolation of Productivity**

Students are constantly connected on Instagram, TikTok, and Telegram with their friends. They're highly social. But when it comes to productivity? That's a solo journey or solo thing. There's no platform that successfully combines the social connectivity students naturally want with the structured accountability they need to actually get things done.

Our survey found (in project proposal) that 73% of students think having friends hold them accountable would help them stay consistent, and 68% are comfortable sharing their habit progress with friends. This tells us there's huge untapped potential for social accountability.

Market Segmentation

Demographic Segmentation

Age Groups

- **17-20 years:** Polytechnic and JC students who are figuring out how to handle increasing responsibilities as they transition into higher education
- **21-25 years:** University undergrads and recent graduates dealing with peak academic demands, internships, and early career decisions

This generation is ambitious and dynamic, but they're also juggling more priorities than ever as they are stepping into their adult life, whether it is school, part-time jobs, extracurriculars, social life, and personal development. The habits they form now will stick with them for years, making this the perfect time to intervene.

Income Levels

- **Allowance-dependent students:** Under S$1,000/month
    - These students are price-sensitive and need affordable or free options
    - They value free features but will engage with in-app reward systems
- **Part-time workers:** S$1,000-2,500/month
    - Students juggling work and studies who really value time management tools
    - Can afford modest subscriptions if the value is clear

Pricing implications: We're planning a freemium model with a solid free tier that includes core features like pact creation, basic verification, and limited consequences. Premium features would be affordable (S$2.99-4.99/month) and include unlimited pacts, priority AI verification, exclusive consequences, and enhanced analytics. We'd also offer student discounts through .edu email verification and potentially partner with universities for campus-wide licenses.

Educational Background

- **Tertiary students:** Polytechnic, JC, and ITE students managing structured curriculum demands
- **University students:** Undergrads handling independent research, project-based learning, and self-directed schedules
- **Postgraduates:** Masters and PhD students who need long-term discipline for thesis work

Psychographic Segmentation

Health & Wellness Consciousness

- **Mental wellness focused:** Students trying to avoid burnout and maintain some balance in their lives
- **Aspirational habit builders:** People who consume tons of productivity content on YouTube and TikTok but struggle to actually implement it
- **Holistic thinkers:** Students who understand that everything's connected, if they're not sleeping or exercising well, their study habits suffer too

These students get that procrastination often isn't about being lazy. It's usually a symptom of stress, anxiety, or burnout. They're looking for tools that approach productivity holistically rather than just pushing "hustle harder" messaging.

Technology Adoption Profile

- **Digital natives:** Gen Z users who uses productivity study tools like Notion, Forest and grew up on Instagram, TikTok, Snapchat, and Telegram
- **Early adopters:** Quick to try new apps, especially if their friends or influencers recommend them
- **Mobile-first users:** They expect smooth, seamless smartphone experiences with minimal friction
- **Privacy-conscious:** Aware of data privacy but willing to share information within trusted friend groups

What this means for design: FoE needs to look and feel as polished as mainstream productivity apps. The learning curve has to be minimal, meaning it is easy to understand. The gamification aspect should also feel rewarding and fun, not gimmicky or punitive.

Lifestyle & Values

- **"Last-Minute Warriors":** Procrastinators who always end up cramming before deadlines, losing sleep, and stressing themselves out
    - They know this pattern is unhealthy but don't have the tools to break it
    - Need external structure and immediate consequences to overcome inertia
- **Social Learners:** Students who thrive in group study settings and enjoy "body doubling" (working alongside others, even if they're not directly collaborating)
    - Motivated by peer perception and social reinforcement
    - Value community features and shared progress tracking
- **Achievement-Oriented:** Students who respond well to streaks, badges, leaderboards, and visible progress metrics
    - Their competitive nature can be channeled productively
    - Enjoy comparing stats with friends in a healthy, non-toxic way

Behavioral Segmentation

Current Productivity Tool Usage

- **To-do list users:** People using Notion, Todoist, Google Tasks, or Apple Reminders
    - Find lists helpful but they lack any enforcement mechanism
    - Often have overflowing task lists that just become sources of guilt
- **Pomodoro practitioners:** Students who've tried Forest, Focus@Will, or similar timer-based apps
    - Appreciate time-blocking but struggle with self-enforcement
    - Looking for more engaging alternatives to simple timers
- **Abandoned app users:** People who've downloaded and quit multiple productivity apps
    - Initial enthusiasm fades within 1-2 weeks
    - Need sustained engagement through social features and variable rewards

Motivation for Using FoE

- **Academic performance:** Preparing for exams, completing assignments on time, maintaining GPA for scholarships
- **Personal discipline:** Building habits for exercise, reading, language learning, skill development
- **Life skills:** Developing consistency with chores, meal prep, budgeting, sleep schedules
- **Career preparation:** LeetCode practice, portfolio building, networking, job applications

Usage Frequency Patterns

- **Daily active users:** Students creating daily pacts for recurring habits like study sessions, exercise, or chores
- **Scenario-based users:** Active during peak stress periods (midterms, finals, project deadlines) but dormant otherwise
- **Streak maintainers:** Users motivated to keep consecutive days of completed pacts going

- **Social users:** Primarily engage by viewing friends' feeds and creating group pacts

Geographic Segmentation

Primary Market: Singapore

- **Major campus hubs:** NUS, NTU, SMU, SIM, SUTD
- **Tertiary institutions:** Polytechnics (SP, NYP, TP, RP, NP) and Junior Colleges across Singapore
- **Urban environment:** High smartphone penetration, reliable internet everywhere, cashless payment infrastructure

Why Singapore works: The high-pressure academic culture, emphasis on meritocracy, and tech-savvy population create ideal conditions for FoE adoption.

Secondary Markets (Future Expansion)

- **Asia-Pacific:** Hong Kong, South Korea, Japan—places with similar academic pressure cultures
- **Western universities:** UK, Australia, US campuses with large international Asian student populations

Primary Target Segment

In order to have more insights and more understanding we purposely chose our segment to have similar characteristic like us, FoE's primary target is:

Demographics:

- Age: 19-23 years old
- Education: University undergraduates in Singapore
- Income: S$800-2,000/month (mix of allowance and part-time work)
- Living situation: Mix of campus housing (halls/residences) and family homes

Psychographics:

- Tech-native and fluent in social media
- Care about peer opinions and social validation
- Want to be productive but can't seem to stay consistent
- Prefer doing things with others rather than alone
- Respond well to gamification and visual progress tracking

Behavioral patterns:

- Active on Instagram, Telegram, and TikTok

- Study in libraries or cafes for the social atmosphere
- Procrastinate by scrolling social media
- Feel guilty about wasted time but keep repeating the same patterns

Pain points:

- Feels guilty after breaking promises to themselves
- Difficulty starting tasks (high activation energy barrier)
- Poor at estimating how long things will actually take
- Constant internal bargaining ("I'll do it later") without real consequences

Needs:

- Immediate accountability that doesn't feel judgmental
- A fun, game-like approach to productivity
- Ability to involve trusted friends
- Clear, visual progress tracking
- Goals that challenge them but still feel achievable

Personas

Persona 1: Chloe Lim - The Socially-Driven Scholar

Demographics

- Age: 21
- Education: Year 2 Business Student (SMU)
- Living: University Hall
- Income: S$1,200/month (S$800 allowance + S$400 part-time retail)
- Tech: iPhone 14, iPad, constantly on Telegram and Instagram

Background

Chloe is outgoing and social—she thrives on interaction but really struggles with self-directed work. She's got a 3.7 GPA, which is good, but she needs 3.8 for her dream consulting internship. She's clearly capable, but she consistently underperforms because she starts assignments way too late.

Goals & Motivations

- Primary: Maintain 3.8 GPA to land a Big 4 consulting internship
- Secondary: Finish work early enough to actually enjoy hanging out at Holland Village with friends guilt-free
- Aspirational: Build a reputation among her friends as someone who has their life together

Pain Points

The "Vague Deadline" Syndrome She sees a deadline two weeks away and thinks "I have time," until suddenly it's 3 AM the night before. She's terrible at estimating how long assignments actually take and optimistically assumes "future Chloe" will be more motivated (spoiler: she never is).

The Social Media Spiral Whenever she hits a difficult part of her work, she opens Instagram or TikTok. "Just 5 minutes" somehow becomes 45 minutes of scrolling. The dopamine hit makes it nearly impossible to maintain focus.

The Guilt-Procrastination Loop She feels guilty about procrastinating, which makes her stressed, which makes her procrastinate more to avoid the stress. She knows she should start earlier but repeats the same pattern with every assignment. Her private to-do lists just make her feel worse without actually changing her behavior.

Behaviors & Preferences

- Only uses apps that look good—design really matters to her
- Heavy Telegram user for group chats, Instagram for social validation
- Studies in cafes or libraries to be "seen" working, but rarely accomplishes deep focus work
- Responds well to social pressure and visible commitment
- Loves sharing achievements but is reluctant to admit failures publicly

Technology Usage

- Screen time: 6.5 hours/day (Instagram 2hrs, TikTok 1.5hrs, Telegram 1hr)
- Has tried: Forest, Notion, Todoist
- Abandoned them because: No social element, felt like nagging, no consequences for just ignoring them

Potential Use of FoE

The Pact: "Complete 1,500 words of Marketing Report" by 9:00 PM tonight

- **Partner:** Her friend Sarah (also a business major, they hold each other accountable)
- **Verification:** Screenshot of word count + AI-generated summary of the content to verify it's not just filler text. As well as her friend confirming that Chloe truly finished her promised task.
- **Consequence if failed:** "The Public Apology"—the app auto-posts a video to her FoE feed where she admits "I let my inner sloth win today. Sarah is superior in productivity and discipline."

Why FoE Works for Chloe: Social stakes make the failure feel real because her friends will actually see it. Pre-commitment removes the internal bargaining she usually does ("but I'm

tired..."). The immediate consequence creates urgency that distant deadlines never provide. Friend verification prevents cheating while maintaining trust with Sarah. The feed-based design feels familiar, like Instagram Stories.

Quote: *"I need my friends to help remind me especially if I don't finish my work. It's the only thing that actually makes me move. Private apps? I just ignore them. If Sarah's going to see me not finishing? That's different."*

Persona 2: Wei Qiang - The Tech-Head Optimizer

Demographics

- Age: 23
- Education: Final Year CS Student (NTU) + Part-time SWE Intern
- Living: Family home in Tampines
- Income: S$2,200/month (S$1,800 internship + S$400 allowance)
- Tech: Custom-built PC, Android flagship, multiple monitors

Background Wei Qiang is analytically minded and loves optimizing systems except he can't seem to optimize his own habits. He's technically skilled but knows that landing a top-tier tech job requires consistent LeetCode practice and project development, not just good coursework. He's been "planning to be consistent" for months but keeps breaking his own commitments.

Goals & Motivations

- Primary: Build a consistent LeetCode habit (1 problem daily) to prepare for FAANG interviews
- Secondary: Reduce his sedentary lifestyle actually he got chronic back pain from poor posture
- Aspirational: Develop a systems-thinking approach to personal productivity that he can share or even teach others

Pain Points

Internal Bargaining Paralysis He rationalizes skipping practice with "one day won't matter in the long run." He's overly analytical about finding the "optimal" practice schedule but never actually executes. His perfectionistic tendencies lead to all-or-nothing thinking.

Weak Immediate Consequences His long-term goal (getting a job offer) feels abstract and far away. There's no immediate penalty for breaking streaks—LeetCode's calendar just shows a gap. He needs tangible, short-term stakes to maintain daily consistency.

Behaviors & Preferences

- Values data visualization, progress charts, and quantified metrics
- Enjoys gaming mechanics like streaks, leveling up, and achievement systems

- Prefers evidence-based approaches over motivational fluff
- Appreciates automation and minimal manual logging
- Responds to systems and incentive structures more than social pressure alone

Technology Usage

- Screen time: 8 hours/day (mostly productive—coding, documentation, Reddit tech forums)
- Has tried: HabitBull, Beeminder, Streaks, Loop Habit Tracker
- Abandoned them because: Beeminder felt too punitive (real money loss), others lacked enforcement, tracking became a tedious chore

Potential Use of FoE

The Pact: "1 hour of LeetCode practice (1 medium problem)"—daily recurring pact

- **Partner:** Jun Wei (fellow CS student also prepping for interviews)
- **Verification:** Screenshot of accepted solution + brief explanation sent to Jun Wei for confirmation
- **Consequence if failed:** "The Fitness Penalty"—submit a video doing 30 push-ups and a 1-minute plank

Secondary Pact: "30 minutes of stretching/exercise"—5 times per week

- **Verification:** Photo of completed routine from fitness app
- **Consequence:** 100 FoE coins redistributed to Jun Wei + "Procrastinator" badge on profile for 24 hours

Why FoE Works for Wei Qiang: It combines accountability (partner review) with data (streak tracking, stats dashboard). The physical consequence for his sedentary habits provides an immediate cost he can actually feel. The coin system gamifies productivity in a measurable way. Public stats and leaderboards appeal to his competitive nature without getting toxic. Partner verification adds a human element that prevents him from gaming the system.

Quote: *"If there's no skin in the game, it's not a commitment—it's just a suggestion. I need progress I can see and consequences I can feel. The LeetCode calendar shows my streak, but who cares if I break it? With FoE, Jun Wei cares. And that makes all the difference."*

Persona 3: Maya Tan - The Overwhelmed Overachiever

Demographics

- Age: 20
- Education: Year 2 Double Degree Student—Business & Communications (NUS)
- Living: Off-campus apartment near Kent Ridge
- Income: S$900/month (family allowance)

- Tech: MacBook Air, iPhone, lives in Google Calendar

Background Maya is ambitious and involved in everything—student government, two CCAs, freelance social media management for small businesses. On paper, she looks like she's crushing it. In reality, she's chronically overwhelmed, constantly behind on coursework, and surviving on 5 hours of sleep. She says yes to everything and follows through on nothing completely.

Goals & Motivations

- Primary: Actually submit assignments on time instead of constantly requesting extensions
- Secondary: Use her carefully crafted Google Calendar instead of ignoring it
- Aspirational: Learn to say no and prioritize ruthlessly

Pain Points

Overcommitment Spiral She agrees to way too many commitments and then can't fulfill any of them properly. She's constantly disappointing others (and herself) with last-minute cancellations. The guilt from breaking promises just compounds her stress.

Planning Without Execution She makes these beautiful, detailed study schedules that she never actually follows. Her Google Calendar is basically aspirational fiction, not reality. The gap between who she plans to be and who she actually is creates constant cognitive dissonance.

Decision Fatigue When you have too many priorities, nothing feels truly urgent. She spends more time reorganizing her task lists than actually doing the tasks. She needs an external forcing function to make her decisions stick.

Behaviors & Preferences

- Color-codes absolutely everything (notebooks, calendars, highlighters)
- Buys productivity planners but abandons them after 2 weeks
- Works best with clear, simple constraints rather than open-ended freedom
- Needs structure imposed from outside, not self-generated

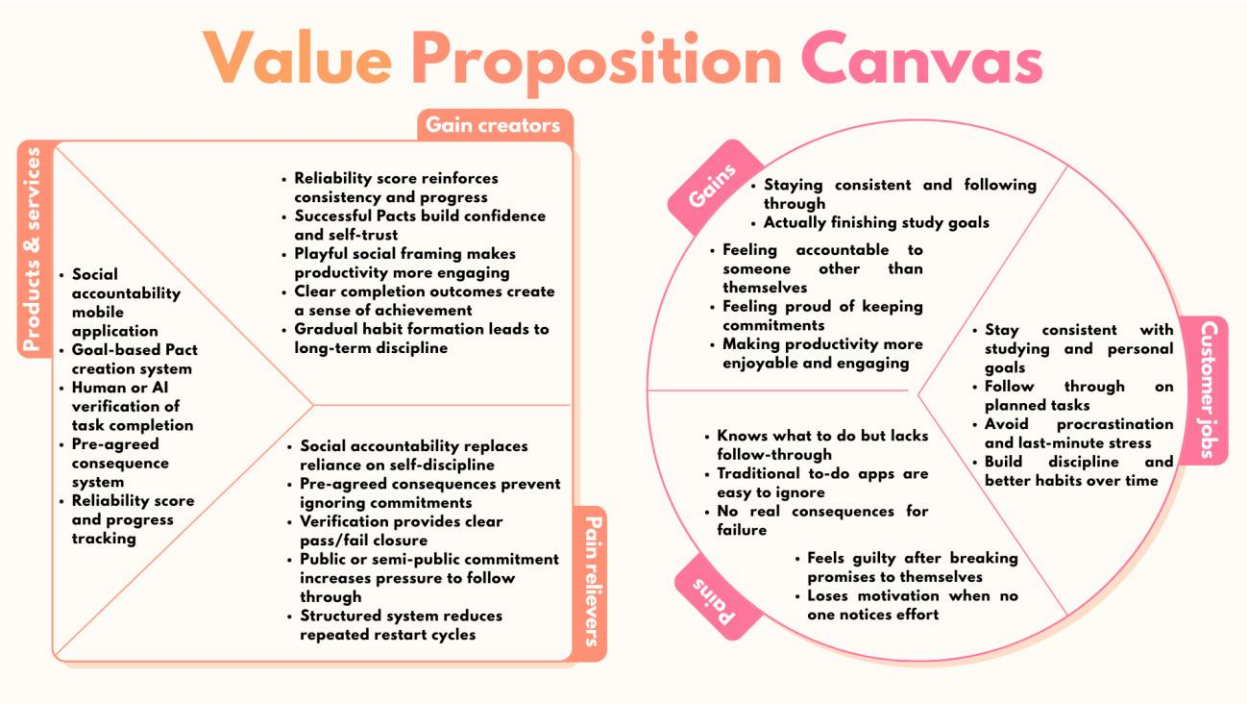
Potential Use of FoE

The Pact: "Finish Comm2101 Essay Draft (1,200 words minimum)" by Friday 11:59 PM

- **Partner:** Study group friend Emma
- **Verification:** Email draft to Emma for a quick skim + AI word count check
- **Consequence:** "The Voucher Freeze" cannot redeem FoE coins for Grab/Foodpanda vouchers for 72 hours (forces her to cook or walk instead of ordering delivery out of convenience)

Why FoE Works for Maya: It forces prioritization—limited active pacts mean she has to choose what actually matters most. External accountability compensates for her weak self-discipline. The consequences create a real cost for overcommitment. Most importantly, she can't just reschedule pact deadlines the way she does with her calendar.

Quote: *"I have a thousand to-do lists and zero follow-through. FoE doesn't let me reschedule my way out of commitments. If I say I'll do it by Friday, it's Friday or consequences. That's the kind of rigidity I need."*



Business Model Canvas

Designed For:

Designed by: Team 2

Date:

Version:

Key Partners

Universities & Student Communities
- Core users

Technology Providers
- AI/ML providers for verification support
- Cloud infrastructure providers

Distribution & Growth Partners
- App stores (Apple App Store, Google Play Store)
- Student organizations and campus ambassadors

Key Activities

Product & System Development
- Developing the mobile app (accountability, verification, consequence system)

Trust, Safety & Moderation
- Content moderation & data safety and security

Platform Operations
- Managing cloud infrastructure and system uptime
- Monitoring performance and scalability

User Growth & Community Engagement
- Student-focused acquisition and referral programs
- Community engagement and retention initiatives

Key Resources

1. Social Trust & Accountability Graph
- Social relationships that give consequences meaning

2. Behavioral & Consequence Effectiveness Data
- Insights into which consequences motivate follow-through

3. Verification & Enforcement System Design
- Pact structure (task, deadline, verification)
- Proof submission flows
- Human + AI verification logic

These resources differentiate FoE from traditional productivity apps.

Value Propositions

- Turning intention into actions through consequences and accountability

Differentiated Features:
- *Social accountability with friends as "Foes" (verifiers)*
- *Gamified experience with coins, streaks, badges, and leaderboard*
- *Privacy-first consequences within app (not external social media)*
- *Instagram-style "For You" feed for friend progress tracking*
- *Pre-defined ethical consequences (social, effort, currency, status, playful)*
- *Blends social connectivity students*

Customer Pain Relief:
- *Eliminates "solo struggle" trap of private habit trackers*
- *Provides immediate stakes for abstract long-term goals*
- *Reduces procrastination through loss aversion and social pressure*

Customer Relationship

Community-Driven:
- Peer-to-peer accountability (friend verification system)
- Social feed fostering engagement and competition

Automated:
- AI verification reduces manual intervention
- Automated notifications for deadlines, friend activity

Channel

Distribution Channels:
- Mobile app stores (Apple App Store, Google Play Store)
- Direct download via website

Marketing/Awareness :
- Social media platforms (Instagram, TikTok, Telegram
- University partnerships and campus ambassador programs
- Word-of-mouth and viral referrals (friend invitations)

Customer Segments

Primary Customer - University Students:
- Full-time tertiary and university students in Singapore (ages 17-25)
- Characteristics: tech-savvy, mobile-first, socially connected, struggle with consistency
- Students who feel they often struggle with procrastinaton & accountability in responsibilities/goals.

Cost Structure

Product Development Costs
- Engineering and design salaries
- Ongoing feature development and maintenance

Administrative & Compliance Costs
- Software tools and service subscriptions
- Legal, accounting, and privacy compliance

Platform & Infrastructure Costs
- Cloud hosting and storage
- AI verification usage (per verification)

Operations & Trust Costs
- Customer support for disputes
- Content moderation and safety enforcement

Monetization & Growth Costs
- Payment processing fees
- Marketing and user acquisition
- Voucher and reward fulfillment

Revenue Stream

Subscription-Based (Primary)
- Freemium tier: Limited active pacts
- Premium subscription:
  - Higher pact limits
  - Advanced verification
  - Customizable consequences
- Student-friendly pricing

Partnership Revenue (Secondary / Controlled):
- Optional brand-sponsored challenges or rewards
- Voucher partnerships capped and controlled, not freely redeemable
- Used primarily for engagement, not cash-out value

Institutional Licenses (Future / B2B)
- University wellness or student success programs
- Corporate productivity pilots